

Caching

CLEANING DATA WITH PYSPARK



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What is caching?

Caching in Spark:

- Stores DataFrames in memory or on disk
- Improves speed on later transformations / actions
- Reduces resource usage

Disadvantages of caching

- Very large data sets may not fit in memory
- Local disk based caching may not be a performance improvement
- Cached objects may not be available

Caching tips

When developing Spark tasks:

- Cache only if you need it
- Try caching DataFrames at various points and determine if your performance improves
- Cache in memory and fast SSD / NVMe storage
- Cache to slow local disk if needed
- Use intermediate files!
- Stop caching objects when finished

Implementing caching

Call `.cache()` on the DataFrame before Action

```
voter_df = spark.read.csv('voter_data.txt.gz')  
voter_df.cache().count()
```

```
voter_df = voter_df.withColumn('ID', monotonically_increasing_id())  
voter_df = voter_df.cache()  
voter_df.show()
```

More cache operations

Check `.is_cached` to determine cache status

```
print(voter_df.is_cached)
```

```
True
```

Call `.unpersist()` when finished with DataFrame

```
voter_df.unpersist()
```

Let's Practice!

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Improve import performance

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Spark clusters

Spark Clusters are made of two types of processes

- Driver process
- Worker processes

Import performance

Important parameters:

- Number of objects (Files, Network locations, etc)
 - More objects better than larger ones
 - Can import via wildcard

```
airport_df = spark.read.csv('airports-*.txt.gz')
```

- General size of objects
 - Spark performs better if objects are of similar size

Schemas

A well-defined schema will drastically improve import performance

- Avoids reading the data multiple times
- Provides validation on import

How to split objects

- Use OS utilities / scripts (split, cut, awk)

```
split -l 10000 -d largefile chunk-
```

- Use custom scripts
- Write out to Parquet

```
df_csv = spark.read.csv('singlelargefile.csv')  
df_csv.write.parquet('data.parquet')  
df = spark.read.parquet('data.parquet')
```

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Cluster sizing tips

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Configuration options

- Spark contains many configuration settings
- These can be modified to match needs
- Reading configuration settings:

```
spark.conf.get(<configuration name>)
```

- Writing configuration settings

```
spark.conf.set(<configuration name>)
```


Cluster Types

Spark deployment options:

- Single node
- Standalone
- Managed
 - YARN
 - Mesos
 - Kubernetes

Driver

- Task assignment
- Result consolidation
- Shared data access

Tips:

- Driver node should have double the memory of the worker
- Fast local storage helpful

Worker

- Runs actual tasks
- Ideally has all code, data, and resources for a given task

Recommendations:

- More worker nodes is often better than larger workers
- Test to find the balance
- Fast local storage extremely useful

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Performance improvements

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Explaining the Spark execution plan

```
voter_df = df.select(df['VOTER NAME']).distinct()
voter_df.explain()
```

```
== Physical Plan ==
*(2) HashAggregate(keys=[VOTER NAME#15], functions=[])
+- Exchange hashpartitioning(VOTER NAME#15, 200)
   +- *(1) HashAggregate(keys=[VOTER NAME#15], functions=[])
      +- *(1) FileScan csv [VOTER NAME#15] Batched: false, Format: CSV, Location:
         InMemoryFileIndex[file:/DallasCouncilVotes.csv.gz],
         PartitionFilters: [], PushedFilters: [],
         ReadSchema: struct<VOTER NAME:string>
```

What is shuffling?

Shuffling refers to moving data around to various workers to complete a task

- Hides complexity from the user
- Can be slow to complete
- Lowers overall throughput
- Is often necessary, but try to minimize

How to limit shuffling?

- Limit use of `.repartition(num_partitions)`
 - Use `.coalesce(num_partitions)` instead
- Use care when calling `.join()`
- Use `.broadcast()`
- May not need to limit it

Broadcasting

Broadcasting:

- Provides a copy of an object to each worker
- Prevents undue / excess communication between nodes
- Can drastically speed up `.join()` operations

Use the `.broadcast(<DataFrame>)` method

```
from pyspark.sql.functions import broadcast
combined_df = df_1.join(broadcast(df_2))
```


Let's practice!
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